

HisabDo AI — Final AI/ML Validation Specification

Financial scoring, spending intelligence, anomaly detection, forecasting, budget recommendations, recommendation rules, accuracy validation, and AI safety

1. Objective

This final AI/ML validation phase closes the HisabDo AI intelligence layer. The purpose is to finalize deterministic financial logic, validate AI/ML behavior against verified backend results, review accuracy and error cases, resolve remaining issues, validate insufficient-data behavior, and prepare the final documentation and demo dataset.

Core rule: the backend/calculation engine is the source of truth for financial values. AI may explain verified results, but it must not invent, alter, or independently calculate financial amounts, scores, percentages, forecasts, or recommendation facts.

2. Final AI/ML Work Items

#	Work Item	Acceptance Evidence
1	Finalize Financial Health scoring logic	Final formula, factor weights, ranges, missing-data rules and sample outputs
2	Finalize Spending Pattern Intelligence	Final trend/category/frequency calculations and thresholds
3	Finalize Anomaly Detection	Final anomaly types, baselines, thresholds and duplicate rules
4	Finalize Cash-Flow Forecasting	Final forecasting method, assumptions, outputs and limitations
5	Finalize Budget Recommendation logic	Final category budget rules, utilization and savings rules
6	Finalize Recommendation Engine rules	Final triggers, priority/severity and conflict handling
7	Review AI/ML validation results	Pass/fail evidence and unresolved issues
8	Verify thresholds and assumptions	Thresholds documented and validated on sample data
9	Verify accuracy metrics	Precision/recall/F1 or feature-appropriate metrics where labels exist
10	Review false positives/false negatives	Cases reviewed, categorized and corrected where required
11	Fix remaining AI/ML issues	Issues retested after fixes
12	Verify insufficient-data behavior	Safe response without fabricated values
13	Verify AI uses verified backend results	DTO/result mapping validated
14	Confirm no financial calculations are invented by AI	No-invention tests passed
15	Prepare final AI/ML documentation	Final methodology, assumptions, metrics and limitations
16	Prepare final AI/ML demo dataset	Controlled, repeatable demo cases
17	Perform final AI validation	End-to-end final AI acceptance

3. Financial Health Scoring — Finalization

- Lock the six approved scoring factors used by the project and document the exact input for each factor.
- Normalize every factor to a consistent 0–100 range before combining factors.
- Document factor weights and ensure the total weight is 100%.
- Calculate the final score only in the backend/calculation engine.
- Define behavior for missing income, expense, budget, debt/receivable, recurring-expense, and cash-flow inputs.
- Validate boundary cases such as zero income, zero expense, no budget, and insufficient history.
- Store the calculation result and factor-level evidence so the AI can explain it without recalculating it.

4. Spending Pattern Intelligence — Finalization

- Finalize top spending categories, frequent expense categories, current-vs-previous period comparisons, and increasing/decreasing trends.
- Use a documented comparison period and minimum-data requirement.
- Document the threshold required before calling a change meaningful.
- Validate category totals against raw transactions.
- Ensure AI receives the verified pattern DTO rather than calculating category totals itself.

5. Anomaly Detection — Finalization

- Finalize Unusual Amount, Sudden Spending Spike, Unusual Category, and Duplicate Transaction detection.
- Document historical baseline, minimum history, deviation threshold and duplicate matching criteria.
- Review false positives such as legitimate one-time large purchases.
- Review false negatives using known anomaly examples.
- Return a structured verified anomaly result containing type, transaction/reference, evidence, severity and confidence where supported.
- AI explains the verified anomaly; it does not decide that an unseen transaction is anomalous.

6. Cash-Flow Forecasting — Finalization

- Finalize the explainable MVP forecasting methodology and input history requirements.
- Forecast expected income, expenses, savings and closing balance using backend calculations.
- Document treatment of recurring transactions and missing months.
- Define minimum history and insufficient-data behavior.
- Validate predicted values against historical/holdout actuals where possible.
- Expose assumptions and limitations with the forecast.

7. Budget Recommendation — Finalization

- Finalize category budget logic using verified income, historical category spending, existing budgets and utilization.
- Define overspending, underspending and savings-opportunity thresholds.
- Validate recommended budgets against sample user histories.
- Ensure recommendations are feasible under the project's documented constraints.
- AI explains the verified budget recommendation and does not generate unsupported amounts.

8. Recommendation Engine — Finalization

- Finalize triggers from Financial Health, Spending Patterns, Anomalies, Cash Flow Forecast, Budget Status and Budget Recommendations.
- Define priority and severity rules.
- Define conflict handling when multiple recommendations are triggered.
- Define duplicate/suppression rules so the same recommendation is not repeatedly shown without a meaningful state change.
- Store reason, action, evidence, priority, severity and verified values in the backend result.

9. Accuracy, False Positives and False Negatives

- Use labeled test cases where ground truth exists.
- For classification/detection tasks, calculate precision, recall and F1 where applicable.
- For forecasting, use appropriate error metrics such as MAE or MAPE only where the data supports them; document the chosen metric and evaluation period.
- For deterministic financial calculations, compare output directly against expected values rather than using ML accuracy metrics.
- Review every false positive and false negative, record root cause, change threshold/rule/model only when justified, and rerun regression tests.
- Do not treat an unlabeled dataset as proof of accuracy.

10. Insufficient-Data and Safety Validation

- No financial history: return an explicit insufficient-data state.
- Partial history: use only supported calculations and disclose limitations.
- Missing category/budget data: do not fabricate categories or budgets.
- Missing forecast inputs: do not invent future income/expense values.
- AI must use unavailable/unknown language instead of guessing.
- All user-scoped backend queries must enforce authorization and data isolation.

11. AI No-Invention Validation

- Send the AI only a verified backend DTO/result.
- Verify every numeric claim in the AI response can be traced to an allowed source field or explicitly supplied calculation result.
- Reject AI output that introduces new amounts, percentages, scores, dates, forecasts or benefits not present in the verified source.
- Reject AI output that changes the recommendation, priority or severity without an explicit backend source.
- For unsupported questions, return a safe limitation instead of a guessed financial answer.

12. Final Demo Dataset

Case	Purpose	Expected Validation
Healthy User	Normal income/expense/budget history	Health, patterns, forecast and recommendations work
High Spending User	Category growth and budget pressure	Spending + budget + recommendation triggers
Anomaly User	Known unusual amount/spike/duplicate	Anomaly detection and explanation
Forecast User	3–6 months history	Forecast values and assumptions
Sparse Data User	Little/no history	Insufficient-data behavior
Multiple Recommendation User	Several triggers at once	Priority/severity and suppression logic

Case	Purpose	Expected Validation
AI Safety User	Verified DTO with controlled values	No-invention validation

13. Final AI Acceptance Checklist

- All financial formulas and deterministic rules finalized.
- All AI/ML thresholds and assumptions documented.
- Accuracy evaluation completed using appropriate metrics.
- False positives reviewed.
- False negatives reviewed.
- Remaining AI/ML issues fixed or explicitly documented.
- Insufficient-data behavior validated.
- Backend result → AI explanation mapping verified.
- No-invention tests passed.
- Final demo dataset prepared.
- Final AI validation completed.
- Final AI/ML documentation completed.